

Single Neuron Computation

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Reversal potential, Resting state and Equilibrium

Electric forces and diffusion are responsible for driving ions across the cell membrane. When a neuron is at its resting state, the current flow due to electric force should cancel the current flow caused by diffusion. What is the membrane potential at the resting state? Can we calculate it specifically? Without losing generality, Let us consider one case for an positive ion (i.e., K^+) with a negative membrane potential. The ion stays inside the cell and the potential outside the cell is higher than that inside. A positive ion inside the cell can cross the membrane only if it has sufficiently large thermal energy to overcome the electrical barrier. In other words, it must have a thermal energy at least $-zeV$ (where $ze > 0$ is the electric charge of the ion, and $V < 0$ is the membrane potential of the neuron. The probability that an ion has thermal energy E follows the Boltzmann distribution $\frac{1}{Z} \exp(-E/k_B T)$, and the probability that the ion can go cross the barrier is simply $\exp(zeV/k_B T)$: this is determined by integrating the Boltzmann distribution for energies $E \geq -zeV$. A concentration of ions inside the cell, n_{in} , that will be able to move across the membrane would be proportional to $n_{in} \exp(zeV/k_B T)$, and this should balance the ions flowing inside the cell, which will be proportional to n_{out} . Putting these things together, we obtain

$$n_{out} = n_{in} \exp(zeE/k_B T). \quad (1)$$

Solving this equation, we have

$$E = \frac{k_B T}{ze} \ln\left(\frac{n_{out}}{n_{in}}\right). \quad (2)$$

Equation 2 is the Nernst equation. The potential we derived is also called the reversal potential: the current flow for a particular type of ion switches its direction when crossing the reversal potential. The reversal potential for a K^+ , denoted as E_K typically falls in the range between -70 and -90 mV; the reversal potential for Na^+ , E_{Na} , is 50 mV or even higher; and E_{Ca} , for Ca^{2+} channels, is even higher, around 150 mV. Cl^- reversal potential are typically around -60 to -65 mV.

The Nernst equation only take into account one type of ion. However, some channels are not quite selective, and we need to combine the current flow from

multiple ions, and the result is the Goldman-Hodgkin-Katz formula for reversal potential. I will write down the equation here, and it is your homework to provide the derivation of this formula.

$$E_m = \frac{k_B T}{e} \ln \left(\frac{\sum_{i=1}^N P_{M_i^+} [M_i^+]_{out} + \sum_{j=1}^N P_{A_j^-} [A_j^-]_{in}}{\sum_{i=1}^N P_{M_i^+} [M_i^+]_{in} + \sum_{j=1}^N P_{A_j^-} [A_j^-]_{out}} \right). \quad (3)$$

When a neuron is not at the resting state, the total current flowing across the membrane through all of its ion channels is called the membrane current of the neuron. By convention, the membrane current is defined as positive when positive ions leave the neuron and negative when positive ions enter the neuron. Let us label different types of channels that may have selective permeability of specific types of ions with index i . As we discussed before, when the membrane potential of a neuron equals to the reversal potential E_i , $V = E_i$, the net current that traverses that channel becomes zero. For many channels, the current increases or decreases linearly with small difference of $V - E_i$. When we add the contribution from different type of ion channels, we have

$$I_m = \sum_i g_i (V - E_i), \quad (4)$$

where g is the ion channel conductances. In Equation 4, we must distinguish two types of conductances. Many ion channels embedded in the membrane are voltage-gated, and therefore the conductances g_i is also voltage-dependent. Some other conductances may be well approximated as voltage independent, such as the current from ion pump, as well as other leaky ion currents. These time-independent conductances could be lumped together by a single term $\bar{g}_L$, and the leaky membrane current I_L is given by:

$$I_L = \bar{g}_L (V - E_L), \quad (5)$$

where E_L is the resting potential of the neuron. Putting everything together, we may write down

$$C_m \frac{dV}{dt} = - \sum_i g_i (V - E_i) - \bar{g}_L (V - E_L) - I_e. \quad (6)$$

Integrate-and-Fire Models

Neurons possess a large repertoire of voltage-gated ion channels that regulate the membrane conductances, making the neuronal activity higher nonlinear. One prominent feature in the activity pattern of many invertebrate and vertebrate neurons is the existence of action potential, or spike. In mammalian brains, action potential is the basic unit for information transmission. Clearly,

the above-mentioned RC circuit cannot be used to describe such nonlinear dynamics. It was in 1952, through a series of elegant experimental and theoretical papers, Hodgkin and Huxley provides a detailed biophysical description of action potential generation and propagation. This work won them the Nobel Prize in 1963.

We will ignore Hodgkin and Huxley, temporarily. On the other hand, neuron models can be simplified and simulations can be accelerated dramatically if the biophysical mechanisms responsible for action potentials are not explicitly included in the model. Integrate-and-fire model (I-F model) is a much simplified, but extremely useful model to describe spike generation. It is widely used in the neuroscience community. In the I-F model, we ignore the voltage dependent term in Equation 6. We only include the linear leaky term together with an *ad-hoc* spiking event. Here the spike is modelled as the point event and it does not have temporal width. The timing of a spike is defined as the time where the membrane potential V reaches the firing threshold value, V_{th} , from below. Whenever a spike occurs the voltage is reset immediately to a lower value, which for simplicity will be taken as $V_{reset} = E_L$. In other words, $V(t_{spike}^-) = V_{th}$, and $V(t_{spike}^+) = V_{reset}$. Now if we define a new variable

$$\tilde{V} = (V - V_{reset}) / (V_{th} - V_{reset}),$$

and denote

$$I_c = g_L(V_{th} - V_{reset}),$$

the RC equation together with the resetting event can be rewritten as

$$\begin{aligned} \tau \frac{d\tilde{V}}{dt} &= -\tilde{V} + \frac{I_e}{I_c} \\ \tilde{V}(t_{spike}^-) &= 1 \\ \tilde{V}(t_{spike}^+) &= 0 \end{aligned} \tag{7}$$

where $I_e > 0$.

To understand the input-output relationship of a neuron, an important quantity that needs to be derived is the dependence of the firing rate, f , on the applied external current I_e . This can be calculated straightforwardly. The time that required to reach the threshold membrane potential is simply given by

$$\frac{I_e}{I_c}(1 - \exp(-t/\tau)) = 1 \tag{8}$$

Rearrange this equation and note that $f = 1/t$, we obtain

$$f = -\frac{1}{\tau \ln(1 - I_c/I_e)}. \tag{9}$$

For large current I_e , using $\ln(1+x) \approx x$, the f-I relationship reduces to

$$f \approx \frac{I_e}{I_c \tau} \quad (10)$$

The firing rate grows linearly with the input current for large I_e .

Now let us examine some experimental data. Real neuron exhibits spike-rate adaptation: the interspike intervals lengthen over time when a constant current is injected into the cell, before a steady-state value is reached. Nevertheless, if one only uses the first two spikes fired by the neuron in response to the injected current, the results agree quite well with the I-F model. The steady-state firing rate can also be fitted by an I-F model, but using a different set of parameters. Another important factor we have not taken into account in our simplest model is the refractory effect: the probability that a neuron fires significantly reduced for a short period of time after the appearance of a spike. This can also be incorporated into the model.

Two-Dimensional Nonlinear Systems

The HH model is difficult to be understood analytically. We thus consider a simplified model

$$C_m \frac{dV}{dt} = -\bar{g}_K n(V - E_K) - \bar{g}_{Na} m_\infty(V)(V - E_{Na}) - \bar{g}_L(V - E_L) + I_e. \quad (11)$$

$$\tau_n \frac{dn}{dt} = n_\infty(V) - n. \quad (12)$$

Note that there are several simplifications we have made. First, we assume that the sodium current is persistent by eliminating the h variable. Second, we drop the exponent in the gating variable of the potassium channel. Third, we shall assume that for a given voltage, the sodium current reaches the steady state instantaneously. In what condition can we make such an assumption?

Before we discuss how to analyze the above two-dimensional system, I would like to briefly introduce some mathematical methods that we will use. This is the minimum requirement you need to understand the lecture.

Nonlinear differential equation

The neural dynamics we deal with are generally nonlinear. The most general differential equation we consider takes the form

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(x) \quad (13)$$

where $\mathbf{x}$ is an N component vector and $\mathbf{f}$ is vector of functions. Unless it is unstable, the behaviors of the dynamics fall into three classes. For one class,

called stable fixed points or point attractors, $\mathbf{x}(t)$ approaches a time-independent vector $\mathbf{x}(\infty)$, when $t \rightarrow \infty$. In another class, called the limit cycle, $x(t)$ becomes periodic at large times and repeats itself indefinitely. For the third class of solutions, the chaotic ones, $x(t)$ never repeats itself but the trajectory of the system lies in a limited subspace of the total space of allowed configurations called a strange attractor. Chaotic solutions are extremely sensitive to initial conditions.

Below we shall consider the point attractor, in which $\mathbf{x}_\infty$ is called the fixed point of the system, when $f(\mathbf{x}_\infty) = 0$. Near the fixed point, we can perform perturbation analysis by expanding f to the first order near the fixed point

$$\mathbf{f}(\mathbf{x}(t)) = \mathbf{f}(\mathbf{x}_\infty) + \mathbf{J}\epsilon(t), \quad (14)$$

where $\mathbf{J}$ is the Jacobian matrix, whose elements are given by

$$J_{ij} = \frac{\partial f_i(\mathbf{x})}{\partial x_j} \quad (15)$$

Thus the dynamics that need to be analyzed near the fixed point is given by

$$\frac{d\epsilon}{dt} = \mathbf{J}\epsilon \quad (16)$$

We know how to analyze this linear equation. The fixed point $\mathbf{x}_\infty$ is a stable fixed point only when the real part of all eigenvalues of the Jacobian matrix $\mathbf{J}$ are negative. Here, let's consider a very simple scenario, where the system we are considering is two-dimensional.

Analysis of a two-dimensional nonlinear system near the fixed point

Let's now consider a very simple case: the Jacobian matrix $\mathbf{J}$ is two-dimensional. In this case, we have

$$\mathbf{J} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The characteristic equation becomes

$$\det \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix} = 0$$

Expanding the determinant yields

$$\lambda^2 - \tau\lambda + \Delta = 0 \quad (17)$$

where $\tau = \text{Tr}\mathbf{J} = a + b$ and $\Delta = \text{Det}\mathbf{J} = ad - bc$. Then we have

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} \quad (18)$$



Figure 1: phase diagram of fixed point stability

Vector field and phase trajectory

A vector function $(x(t), y(t))$ is a solution of the two-dimensional system

$$\dot{x} = f(x, y), \quad (19)$$

$$\dot{y} = g(x, y), \quad (20)$$

starting with an initial condition $(x(0), y(0)) = (x_0, y_0)$. This requirement has a simple geometrical interpretation: a solution is a curve $(x(t), y(t))$ on the phase plane which is tangent to the vector field.

Poincare-Bendixson Theorem

See slides.

Now let us go back to the two-dimensional model. Let us determine the nullclines of the $I_{Na,p} + I_K$ -model. The V-nullcline is given by the equation

$$I_e - \bar{g}_L(V - E_L) - \bar{g}_{Na}m_\infty(V)(V - E_{Na}) - \bar{g}_Kn(V - E_K) = 0 \quad (21)$$

which has the solution

$$n = \frac{I_e - \bar{g}_L(V - E_L) - \bar{g}_{Na}m_\infty(V)(V - E_{Na})}{\bar{g}_K(V - E_K)} \quad (22)$$

It typically has the form of a cubic parabola. The steady state of the gating variable n defines the n -nullcline, given by

$$n = n_\infty(V) \quad (23)$$

The V - and n -nullclines partition the phase plane into four regions, in each of which the vector field has a different direction:

- (a) Both V and n increase.
- (b) V decreases but n still increases.
- (c) Both V and n decrease.
- (d) V increases but n still decreases.